

# Chain of Spatial Thoughts: Modality-Agnostic Spatial Grounding for Vision Language Models

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## Abstract

Spatial understanding is fundamental to embodied intelligence, underpinning applications such as robotic manipulation, embodied navigation, and autonomous driving. Although recent vision-language models (VLMs) have achieved impressive performance on spatial reasoning benchmarks, state-of-the-art approaches typically rely on additional spatial encoders or architectural modifications during inference, increasing computational cost. We introduce **Space Tokens**, a lightweight, architecture-agnostic framework that equips VLMs with explicit continuous spatial representations without requiring additional inference-time modules. By distilling scene-level 3D geometry and object-centric spatial attributes into continuous latent tokens, our method enables these modalities to be directly incorporated into a chain-of-thought reasoning process, thereby improving the VLM’s spatial reasoning capabilities. At the same time, the learned representations can be explicitly decoded to verify that they encode meaningful geometric information, while the unified token interface remains extensible to additional modalities. Experiments on VSI-Bench improve Qwen3-VL-8B by 4.3% and SenseNova-SI-1.3 by 1.3%, while achieving state-of-the-art performance on object size (79.2%) and room size estimation (75.7%). These results demonstrate that continuous spatial tokens provide an effective, interpretable, and computationally efficient mechanism for integrating geometric reasoning into large vision-language models.

## Introduction

Recent advances in Multimodal Large Language Models (MLLMs), also referred to as VLMs, have enabled systems that jointly reason over visual observations and natural language (Qwen Team 2025; Li et al. 2023; Deng et al. 2025; Gemini Team 2025; Gemma Team 2025). Beyond traditional image understanding, there is growing interest in equipping these models with the ability to understand the underlying 3D geometry, spatial relationships, and physical interactions depicted in images and videos (Yang et al. 2025; Fan et al. 2026). Such spatial intelligence is fundamental to real-world applications including robotics, embodied agents, autonomous driving, and human-computer interaction, where successful decision making depends not only on recognizing objects, but also on understanding their geometric arrangement and interactions within the surrounding environment.

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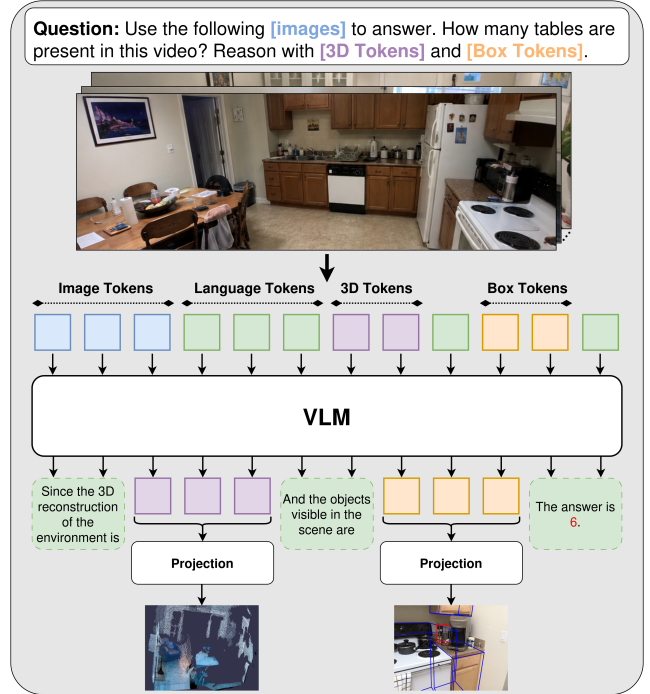


Figure 1: Space Tokens Overview. Our approach embeds interpretable scene-level and object-centric spatial representations directly into the VLM for use during autoregressive reasoning. The learned tokens improve spatial reasoning and can be decoded into explicit spatial modalities to verify the model’s learned geometric understanding

Progress in spatial reasoning has largely followed two complementary directions. First (1), *foundational scaling* enhances spatial capabilities as a byproduct of expanding model parameter quantity and training datasets with internet-scale data (Qwen Team 2025; Deng et al. 2025; Gemini Team 2025) or by training on increasingly large spatially focused supervised datasets, as demonstrated by recent models such as SenseNova-SI (Cai et al. 2026). Second (2), *specialized spatial architectures*, introduces explicit spatial inductive biases through dedicated geometry modules, including 3D reconstruction encoders, geometry-aware connectors, object-

centric representations, and scene-level relational reasoning (Zhao et al. 2025; Li et al. 2026; Chen et al. 2026b; Wang et al. 2025b; Liu et al. 2025; Zhang et al. 2026). More recently, Latent Token-Based Knowledge Integration has emerged as a general methodology for incorporating auxiliary knowledge into VLMs through continuous latent token representations, allowing external representations to be distilled into the model while preserving the underlying architecture (Qin et al. 2025; Yang et al. 2026; Ray et al. 2026). Existing works have primarily leveraged this paradigm to internalize visual representations for improving perception and general 2D reasoning, leaving its potential for encoding explicit spatial and geometric knowledge largely unexplored.

Together, these approaches have advanced performance on benchmarks such as VSI-Bench (Yang et al. 2025), VLM-3R (Fan et al. 2026), and EASI (Cai et al. 2025), yet key limitations remain. Scaling-based methods require large models or extensive spatial supervision, while explicit spatial approaches add inference-time modules that increase cost and reduce portability. Latent token methods avoid these drawbacks but have largely focused on visual or semantic knowledge, leaving explicit 3D geometry and object-centric representations underexplored. As a result, current VLMs still lag behind human spatial understanding, particularly on long-horizon tasks such as route planning and object appearance ordering (Yang et al. 2025).

In this work, we build upon the paradigm of Latent Token-Based Knowledge Integration and extend it to spatial intelligence. Rather than scaling model capacity, scaling spatial supervision, or introducing large-size specialized inference-time geometry modules, we investigate whether explicit 3D geometric and object-centric spatial knowledge can instead be internalized within the existing VLM through lightweight training. As illustrated in Fig. 1, we introduce **Space Tokens**, a model-agnostic framework that distills scene-level geometry and object-centric spatial representations into continuous latent tokens that become part of the VLM’s native chain-of-thought reasoning process. By extending latent token representations from visual understanding to explicit spatial knowledge, our method improves spatial reasoning while preserving the efficiency and portability of the underlying VLM. Furthermore, the learned representations remain directly decodable into interpretable 3D modalities, enabling explicit verification that the model has acquired meaningful geometric understanding without requiring architectural modifications or additional inference-time modules.

Our main contributions are summarized as follows:

- We introduce Space Tokens, a modality-agnostic and visually verifiable method that equips VLMs with continuous spatial representations through a unified token interface during chain-of-thought reasoning, requiring no additional inference-time modules or architecture modifications.
- We improve scene-level 3D understanding by distilling geometric knowledge from large 3D reconstruction models directly into the VLM’s latent space, removing the need for these models at inference time.
- We also enhance object-centric spatial reasoning by en-

coding 3D bounding-box information, including object position, orientation, and size, into dedicated latent tokens.

- We demonstrate the effectiveness of Space Tokens on several spatial, visual and general VQA benchmarks. Our method improves Qwen3-VL-8B by 4.3 points and SenseNova1.3 by 1.3 points on VSI-Bench while achieving state-of-the-art performance on object size and room size estimation.

## Related Work

**Spatial Foundational Scaling.** Recent improvements in spatial reasoning have been driven in part by scaling the capacity of general-purpose vision-language models and the amount of spatially supervised training data. Large multi-modal foundation models such as Qwen3-VL, Gemini, and BAGEL demonstrate strong zero-shot and few-shot spatial reasoning capabilities that emerge from large-scale multi-modal pretraining (Qwen Team 2025; Deng et al. 2025; Gemini Team 2025). Complementing model scaling, SenseNova-SI (Cai et al. 2026) shows that scaling high-quality spatial instruction data substantially improves spatial reasoning across a wide range of tasks. Although these approaches establish increasingly capable general-purpose models, they typically require substantially larger models or datasets to achieve continued improvements.

**Explicit Spatial Representations.** A second line of work improves spatial reasoning by explicitly introducing geometric representations into the VLM. Representative approaches include augmenting the VLM with dedicated geometry modules such as the camera-guided fusion architecture of SpaceMind (Zhao et al. 2025), geometry encoders in Geo-Thinker and VG-LLM (Li et al. 2026; Zheng et al. 2025), and reconstruction-guided reasoning frameworks such as VLM-3R (Fan et al. 2026). Beyond scene-level geometry, several methods incorporate object-centric spatial representations. LocateAnything3D (Man et al. 2025) formulates 3D object detection as next-token prediction through a Chain-of-Sight procedure, while N3D-LM (Wang et al. 2025b) jointly learns 3D grounding and spatial reasoning using 3D-aware QA supervision. SandboxVLM (Liu et al. 2025) represents scenes through abstract 3D bounding boxes, and SSR (Zhang et al. 2026) combines object-centric 7D representations with lightweight scene graphs. Chen et al. (Chen et al. 2026a) further unify 6D object parsing, tracking, and geometric prediction before reinforcement learning refinement. Collectively, these methods demonstrate that explicit geometric representations significantly improve spatial reasoning; however, they generally rely on specialized architectural components or external geometry modules that remain active during inference, increasing computational cost and reducing portability across VLM backbones.

**Latent Token-Based Knowledge Integration.** Rather than introducing additional architectural components, recent works have explored continuous latent tokens as a general mechanism for incorporating auxiliary knowledge into VLMs. Mirage (Yang et al. 2026) introduces latent visual

tokens that represent the model’s internal mental imagery, enabling reasoning over visual concepts without explicitly generating intermediate images.

Chain-of-Visual-Thought (CoVT) (Qin et al. 2025) instead focuses on enriching visual perception by distilling complementary information from multiple pretrained expert models, including depth estimation, semantic segmentation, and edge detection, into continuous latent representations that improve fine-grained image understanding. Collectively, these works establish latent tokens as an effective paradigm for internalizing auxiliary knowledge while preserving the underlying VLM architecture. However, existing approaches primarily employ latent tokens to encode visual representations that support perception and general 2D reasoning, leaving the incorporation of explicit 3D geometric and object-centric spatial knowledge largely unexplored.

Our work extends latent token-based knowledge integration from visual perception to explicit spatial intelligence. We embed complementary scene-level and object-centric 3D information into continuous latent tokens, enabling VLMs to internalize spatial priors within their native autoregressive reasoning process without architectural changes or additional inference-time modules. More broadly, latent tokens provide a unified interface for integrating structured spatial knowledge into foundation VLMs.

## Methodology

### Continuous Spatial Tokens

To preserve the underlying autoregressive generation paradigm of the VLM and ensure that our approach remains lightweight and model agnostic, we introduce spatial tokens by reserving a subset of the vocabulary, without requiring any architectural modifications. Given a set of images or video frames  $\mathcal{I} = \{I_1, I_2, \dots, I_n\}$ , a prompt  $\mathcal{P}$ , and model  $M_\theta$ , the output sequence  $\mathcal{Y}$  is generated according to

$$M_\theta(\mathcal{Y}|\mathcal{I}, \mathcal{P}) = \prod_{t=1}^{|\mathcal{Y}|} M_\theta(y_t|\mathcal{I}, \mathcal{P}, y_{<t}) \quad (1)$$

During autoregressive generation, spatial tokens are produced identically to any other tokens. Rather than using their discrete token identities, we extract the final layer hidden states corresponding to the generated positions of the spatial tokens. These serve as continuous spatial representations for downstream supervision.

Based on these representations, we define continuous spatial tokens that encode specific spatial concepts such as scene-level 3D reconstruction and instance-level 3D bounding boxes. Unlike conventional visual tokens that primarily capture semantic information, these tokens are optimized to represent fine-grained spatial structure enabling the VLM to reason about the geometry of its environment. To learn these representations, we supervise the spatial tokens using task-specific objectives. When a teacher model is available, spatial tokens are aligned with the teacher through feature-level and representation-level objectives, similar to CoVT (Qin et al. 2025). For spatial concepts without an appropri-

ate teacher model, spatial tokens are learned solely through the supervised task objective.

**3D Representation Tokens** We align the 3D representation tokens with the latent scene representation learned by VGGT-Omega (Wang et al. 2025a). The objective is twofold. First, we encourage the VLM latent tokens to encode the same geometric information as the teacher through latent space alignment. Second, we ensure the representations remain geometrically meaningful by supervising their decoded 3D outputs. Accordingly, the 3D representation tokens are jointly optimized with similarity and multi-task reconstruction losses. The latent space objective encourages the continuous spatial tokens to capture the teachers internal geometric representation, while the reconstruction objectives ensure that these representations remain predictive of explicit 3D scene structure.

For latent space alignment, we project the hidden states corresponding to the 3D reconstruction tokens into the VGGT-Omega latent space,  $\hat{z} = \text{proj}(h_{3D})$ , where  $h_{3D}$  denotes the final-layer hidden states of the VLM backbone associated with the output 3D tokens. We then minimize the cosine distance between the projected representation,  $\hat{z}$ , and the corresponding VGGT-Omega aggregation features,  $z$ :

$$\mathcal{L}_{sim} = 1 - \frac{\hat{z}^\top z}{\|\hat{z}\| \|z\|} \quad (2)$$

To further encourage the latent tokens to encode complete scene geometry, we additionally supervise the decoded outputs using the reconstruction objectives proposed by VGGT-Omega. Specifically, we reconstruct camera parameters, dense depth maps, and 3D point maps. The camera reconstruction objective,  $\mathcal{L}_{cam} = \sum_{i=1}^N |\hat{\mathbf{g}}_i - \mathbf{g}_i|$ , is an  $\ell_1$ -objective between the predicted camera pose encoding vector,  $\hat{\mathbf{g}}_i$ , and the teacher estimation,  $\mathbf{g}_i$ . Following the original VGGT implementation (Wang et al. 2025a), the depth reconstruction objective combines a regression term with a gradient consistency term, both weighted by an aleatoric uncertainty map,  $\mathcal{L}_{dpt} = \sum_{i=1}^N \|c_i^D \odot (\hat{D}_i - D_i)\| + \|c_i^D \odot \nabla (\hat{D}_i - D_i)\| - \alpha \log(c_i^D)$ , where  $D_i$  and  $\hat{D}_i$  denote the teacher and predicted depth maps, respectively, and  $c_i^D$  represents the predicted uncertainty. The predicted depth maps are unprojected into camera coordinates using the estimated camera intrinsics to obtain  $\hat{P}$ , following the formulation from VGGT-Omega. The resulting point maps are supervised analogously to the depth objective,  $\mathcal{L}_{pt} = \sum_{i=1}^N \|c_i^D \odot (\hat{P}_i - P_i)\| + \|c_i^D \odot \nabla (\hat{P}_i - P_i)\|$  except that the aleatoric uncertainty term is omitted, as its effect is already accounted for in the depth supervision. Additional methodological details of the 3D reconstruction objective formulation are provided in the appendix.

The complete training objective for the 3D reconstruction tokens is:

$$\mathcal{L}_{3D} = \lambda_{sim} \mathcal{L}_{sim} + \lambda_{cam} \mathcal{L}_{cam} + \lambda_{dpt} \mathcal{L}_{dpt} + \lambda_{pt} \mathcal{L}_{pt} \quad (3)$$

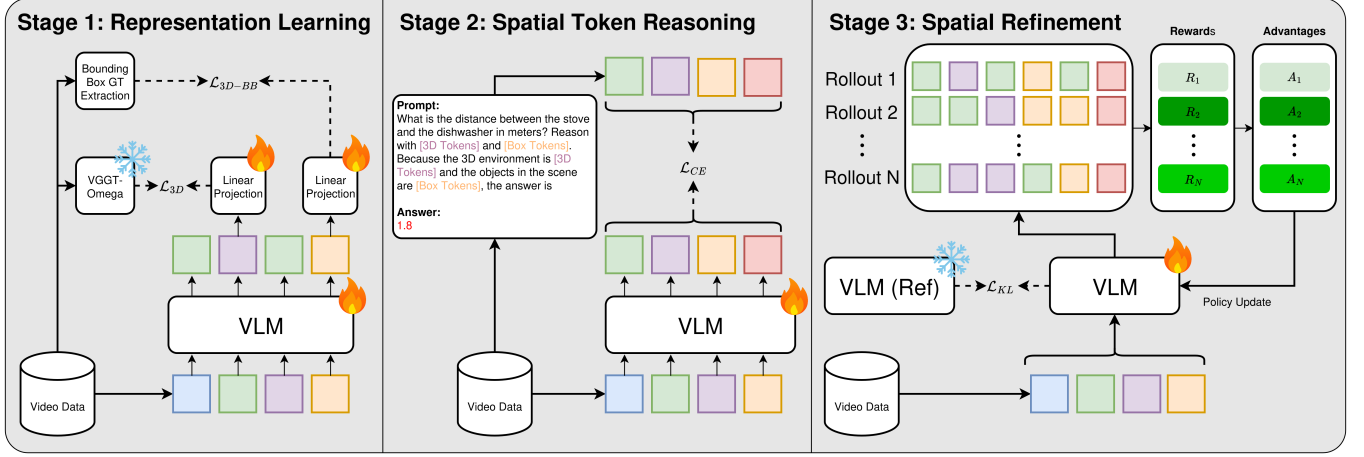


Figure 2: Overview of the proposed three-stage Space Tokens training pipeline. **Stage 1** learns continuous spatial token representations by distilling scene-level 3D geometry from VGGT-Omega and object-centric spatial attributes into reserved output vocabulary tokens. **Stage 2** supervises the VLM to reason with the learned spatial tokens through standard next-token prediction SFT. **Stage 3** further refines spatial reasoning using reinforcement learning. After stage 1, all spatial reasoning is performed using the learned latent tokens without requiring additional model parameters.

where the  $\lambda$  coefficients balance the contribution of each objective. The values used in our experiments are provided in the appendix.

**3D Bounding Box Tokens** To enhance the model’s object-centric understanding of the environment, we introduce a set of dedicated tokens for predicting object-level 3D bounding boxes. We adopt this representation because a 3D bounding box compactly encodes an object’s position, orientation, and spatial extent. These properties provide useful geometric priors for answering questions involving object dimensions, relative poses, spatial relationships, and scene layout.

To predict 3D bounding boxes, we combine the hidden representations of the dedicated bounding-box tokens with visual features extracted by the VLM’s vision backbone. We additionally incorporate features derived from the corresponding normalized 2D bounding boxes to provide object-localization guidance during training. As discussed previously, the auxiliary modalities do not necessarily need to be decoded at inference time. Consequently, the 2D bounding boxes are used only as training-time supervision and are not required during inference.

We formulate 3D bounding-box prediction as fixed-cardinality set regression. Each box is represented by a 12D vector containing its camera-space center, dimensions, and 6D rotation representation. Predictions are matched to ground-truth boxes using Hungarian assignment with an  $\ell_1$  matching cost. The bounding-box objective is defined as

$$\mathcal{L}_{3D-BB} = \lambda_{obj}\mathcal{L}_{obj} + \lambda_{bg}\mathcal{L}_{bg}, \quad (4)$$

where  $\mathcal{L}_{obj}$  is the Smooth  $\ell_1$  loss ( $\beta = 0.1$ ) over Hungarian-matched boxes, and  $\mathcal{L}_{bg}$  regresses unmatched slots toward a zero background target. The loss is summed over prediction slots and averaged over annotated views. The complete decoder architecture and objective are provided in the appendix.

### Three-Stage Pipeline

The full training pipeline of the Space Tokens method is illustrated in Fig. 2. To adequately learn both the spatial representations and how to effectively use these representations during the reasoning process, we design a three-stage training pipeline. The first stage is the representation learning stage, in this stage we manually inject the spatial tokens in the prompt, directly embedded within a chain-of-thought style reasoning sentence. We compute the alignment losses using Eq. 3 and Eq. 4 by projecting the hidden states associated with the spatial tokens into their corresponding latent representations. In this stage, we only backpropagate these reconstruction losses to ensure that the projection layers learn a suitable mapping from the VLM hidden state to the spatial feature latent space, without having to compete with downstream tasks.

The second stage is the spatial-token reasoning stage. During this stage, we freeze the projection layers that map the VLM hidden states to the spatial-feature latent space, preventing the learned spatial representations from drifting during training. As in the first stage, the spatial tokens are retained in the prompt. We additionally optimize a cross-entropy objective under teacher forcing to encourage the VLM to explicitly incorporate the spatial tokens into its reasoning process. Importantly, our empirical results indicate that reasoning over a subset of the input frames is more effective than reasoning over the full set. Additional details regarding the prompt structure and frame-selection strategy are provided in the appendix. We also retain the spatial-alignment losses during this stage as regularization objectives. This helps preserve the spatial representations learned in the first stage while the model is optimized to produce the correct final answer.

The final stage is a spatial refinement stage. We perform group relative policy optimization (GRPO) (Shao et al. 2024) to further improve the ability of the model to reason with spa-



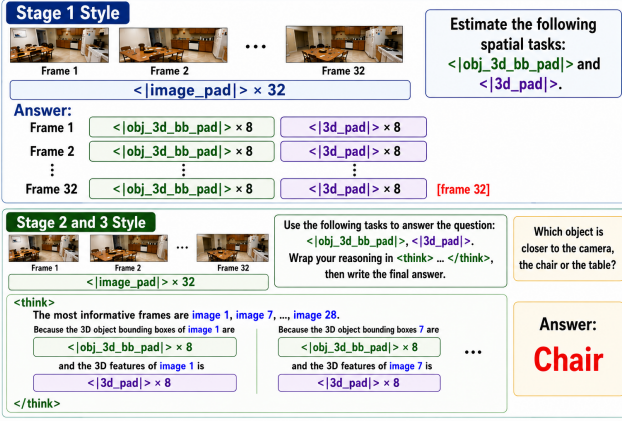


Figure 3: Prompt formats used across training stages. Top (Stage 1): We perform representation learning by training the dedicated slots to encode spatial information. Bottom (Stages 2 and 3): We leverage the learned representations to improve the model’s spatial reasoning, first through SFT and then through GRPO trainings.

tial representations. For a image set and prompt pair,  $(\mathcal{I}, \mathcal{P})$ , we generate a group of  $G$  output sequences,  $\{\mathcal{Y}_1, \dots, \mathcal{Y}_G\}$  from the current model  $M_{\theta_{old}}$ . Using a scoring function to determine a reward for each sequence, we can take a policy gradient step to update  $M_{\theta_{old}}$  to  $M_{\theta}$  by optimizing the following cost

$$\mathcal{J}(\theta) = \frac{1}{G} \sum_{i=1}^G \frac{1}{|\mathcal{Y}_i|} \sum_{t=1}^{|\mathcal{Y}_i|} \left\{ \min \left[ \text{clip} \left( r_{i,t} \hat{A}_{i,t}, 1 - \epsilon, 1 + \epsilon \right), r_{i,t} \hat{A}_{i,t} \right] - \beta \mathbb{D}_{KL} \left[ M_{\theta} || M_{\text{ref}} \right] \right\} \quad (5)$$

where  $r_{i,t} = \frac{M_{\theta}(y_{i,t} | y_{i,<t})}{M_{\theta_{old}}(y_{i,t} | y_{i,<t})}$  is the likelihood ratio for sampling the token from sequence  $\mathcal{Y}_i$  at step  $t$ , and  $\hat{A}_{i,t} = \frac{R_{i,t} - R_t}{\sigma_R}$  is the group-normalized advantage for some reward,  $R_t$ .  $\mathbb{D}_{KL}$  represents the Kullback-Leibler divergence which penalizes the model from diverging from the stage 2 reference model, and  $\epsilon$  and  $\beta$  are hyperparameters that control the magnitude of policy deviation and influence of the KL divergence penalty, respectively. Details on the reward design are included in the appendix.

## Experiments

### Implementation Details

**Training Setup** Space Tokens models are fine-tuned on top of the pretrained Qwen3-VL-8B and SenseNova-SI-1.3 VLMs, applying low-rank adaption (LoRA) (Hu et al. 2021) with rank 16 and scaling factor of 32. Before performing GRPO, the LoRA adapter from the prior SFT stages is merged, and a new LoRA adapter with the same configuration is instantiated. A cosine learning rate scheduler is employed using a learning rate of  $5 \times 10^{-5}$  and  $1 \times 10^{-6}$  for

the SFT and GRPO stages, respectively. More detailed training configurations are described in the appendix.

**Data** The Space Tokens models are fine-tuned for 1 epoch using the VICA-322K (Feng 2025) dataset for both stage 1 & 2, and stage 3 exposes the model to an additional 8,000 samples. VICA-322K is chosen because it provides large-scale instructional video sequences containing diverse environments and objects, making it well-suited for learning continuous scene-level geometry and object-centric spatial relationships. During data processing, 32 images are uniformly sampled from the input video and are resized to  $448 \times 448$  pixels. Further details on dataset configuration and processing are provided in the appendix.

**Benchmark** We mainly evaluate Space Tokens on VSI-Bench (Yang et al. 2025), which contains over 5,000 question and answer pairs designed to evaluate spatial understanding in indoor environments, with data curated from ARKitScenes (Dehghan et al. 2021), ScanNet (Dai et al. 2017), and ScanNet++ (Yeshwanth et al. 2023). VSI-Bench considers two categories of questions, multiple choice (MCA) and numerical answer (NA), which span eight tasks: object counting, absolute distance, object size, room size, relative distance, relative direction, route planning, and object appearance order. Evaluation on VSI-Bench is conducted following the official protocol, and uses mean accuracy for MCA and relative accuracy for NA.

To verify that our training procedure does not induce catastrophic forgetting or degrade the VLM’s general capabilities, we also evaluate the resulting models on several related but out-of-distribution (OOD) benchmarks. Specifically, BLINK (Fu et al. 2024) and CV-Bench (Tong et al. 2024) assess visual perception and spatial understanding, while NEXT-QA (Xiao et al. 2021) assesses general video-based question answering.

### Quantitative Results

Table 1 reports the results on the full VSI-Bench benchmark. We compare Space Tokens against proprietary VLMs, open-source general-purpose VLMs, architecture-customized spatial reasoning models, and prior training-only approaches. Across both the pretrained Qwen3-VL-8B and stronger SenseNova-SI-1.3 checkpoints, Space Tokens consistently improves spatial reasoning while leaving the underlying VLM architecture unchanged. Relative to the pretrained Qwen3-VL-8B model, our method improves the overall VSI-Bench score by 4.3 percentage points, with the largest gains observed on room size (+15.8) and absolute distance (+5.1). Similar trends are observed when applied to the stronger SenseNova-SI-1.3 checkpoint, where stage 3 further improves the overall score by 1.3 percentage points, with largest gains observed on room size (+9.5). Space Tokens models achieve state-of-the-art performance on object size, and room size tasks, and obtain competitive performance across remaining tasks, including second-best results on object counting and appearance order. The substantial improvement in room size estimation is particularly notable, as this task depends heavily on understanding the global 3D geometry of

Table 1: Results on VSI-Bench full split. Methods are grouped by model type. Best results are shown in **bold**, second best results are underlined. “–” indicates results not reported. Space Tokens variants are highlighted **blue** and the pretrained checkpoints are highlighted **orange**.

| Method                                            | Avg.  | Numerical Questions |            |           |           | Multiple-Choice Questions |           |            |             |
|---------------------------------------------------|-------|---------------------|------------|-----------|-----------|---------------------------|-----------|------------|-------------|
|                                                   |       | Obj. Cnt.           | Abs. Dist. | Obj. Size | Room Size | Rel. Dist.                | Rel. Dir. | Route Plan | Appr. Order |
| Proprietary Models                                |       |                     |            |           |           |                           |           |            |             |
| GPT-4o (GPT Team 2024)                            | 34.0  | 46.2                | 5.3        | 43.8      | 38.2      | 37.0                      | 41.3      | 31.5       | 28.5        |
| Gemini-1.5 Flash (Gemini Team 2025)               | 42.1  | 49.8                | 30.8       | 53.5      | 54.4      | 37.7                      | 41.0      | 31.5       | 37.8        |
| Gemini-1.5 Pro                                    | 45.4  | 56.2                | 30.9       | 64.1      | 43.6      | 51.3                      | 46.3      | 36.0       | 34.6        |
| Gemini 3 Pro                                      | 52.5  | 38.0                | 37.8       | 72.7      | 44.1      | 59.9                      | 55.7      | 45.9       | 66.0        |
| GPT-5.2                                           | 54.5  | —                   | —          | —         | —         | —                         | —         | —          | —           |
| Open-Source General VLMs                          |       |                     |            |           |           |                           |           |            |             |
| LLaVA-Next-Video-7B                               | 35.6  | 48.5                | 14.0       | 47.8      | 24.2      | 43.5                      | 42.4      | 34.0       | 30.6        |
| LLaVA-OneVision-72B (Li et al. 2024)              | 40.9  | 43.5                | 23.9       | 57.6      | 37.5      | 42.5                      | 39.9      | 32.5       | 44.6        |
| InternVL3-78B (InternVL Team 2025)                | 48.5  | 71.2                | 53.7       | 44.4      | 39.5      | 55.9                      | 39.5      | 28.9       | 54.5        |
| Qwen3-VL-8B (Qwen Team 2025)                      | 59.8  | 67.1                | 52.3       | 76.1      | 56.7      | 59.9                      | 54.1      | 38.7       | 73.3        |
| Architecture-Customized Spatial Models            |       |                     |            |           |           |                           |           |            |             |
| Spatial-MLLM (Wu et al. 2025a)                    | 48.4  | 65.3                | 34.8       | 63.1      | 45.1      | 41.3                      | 46.2      | 33.5       | 46.3        |
| VLM-3R (Fan et al. 2026)                          | 60.9  | 70.2                | 49.4       | 69.2      | 67.1      | 65.4                      | 80.5      | 45.4       | 40.1        |
| SpaceMind (Zhao et al. 2025)                      | 69.6  | 73.3                | 61.4       | 77.3      | 74.2      | 67.2                      | 88.4      | 44.3       | 70.6        |
| GeoThinker (Li et al. 2026)                       | 72.6  | 74.6                | 62.8       | 76.9      | 74.9      | 75.9                      | 90.2      | 48.4       | 77.5        |
| Training-Only (No Architecture Modification)      |       |                     |            |           |           |                           |           |            |             |
| SpaceR (Ouyang et al. 2025)                       | 45.5  | 57.8                | 28.2       | 59.9      | 47.1      | 40.1                      | 45.4      | 33.5       | 52.1        |
| ViLaSR (Wu et al. 2025b)                          | 45.4  | 63.5                | 34.4       | 60.6      | 30.9      | 48.9                      | 45.2      | 30.4       | 49.2        |
| SenseNova-SI-1.2 (InternVL3-8B) (Cai et al. 2026) | 69.6  | 72.7                | 56.0       | 77.1      | 75.7      | 70.4                      | 81.7      | 42.8       | 79.9        |
| SenseNova-SI-1.3 (Qwen3-VL-8B) <sup>1</sup>       | 67.6  | 72.7                | 54.9       | 77.1      | 66.2      | 70.1                      | 81.0      | 42.3       | 76.4        |
| Space Tokens                                      |       |                     |            |           |           |                           |           |            |             |
| Qwen3-VL-8B                                       | 64.1  | 74.4                | 57.4       | 77.8      | 72.5      | 64.4                      | 55.1      | 37.6       | 73.5        |
| Δ Improvement                                     | ↑ 4.3 | ↑ 7.3               | ↑ 5.1      | ↑ 1.7     | ↑ 15.8    | ↑ 4.5                     | ↑ 1.0     | ↓ 1.1      | ↑ 0.2       |
| SenseNova-SI-1.3 (Qwen3-VL-8B)                    | 68.9  | 71.8                | 59.4       | 79.2      | 75.7      | 68.2                      | 77.6      | 41.8       | 77.8        |
| Δ Improvement                                     | ↑ 1.3 | ↓ 0.9               | ↑ 4.5      | ↑ 2.1     | ↑ 9.5     | ↓ 1.9                     | ↓ 3.4     | ↓ 0.5      | ↑ 1.4       |
| Human                                             | 79.2  | 94.3                | 47.0       | 60.4      | 45.9      | 94.7                      | 95.8      | 95.8       | 100         |

Table 2: Results on OOD benchmarks. We evaluate the trained model on related but unseen benchmarks to assess whether it preserves its general capabilities beyond the training tasks. BLINK and CV-Bench evaluate visual perception and spatial understanding, while NExT-QA measures general video-based question answering. The results show that our method maintains, and in some cases improves, performance across these tasks.

| Method                                   | BLINK | CV-Bench | NExT-QA |
|------------------------------------------|-------|----------|---------|
| Qwen3-VL-8B + SenseNova-SI               | 64.4  | 88.4     | 74.4    |
| Qwen3-VL-8B + SenseNova-SI + Stage 1 & 2 | 64.9  | 86.9     | 80.0    |
| △ Improvement                            | ↑ 0.5 | ↓ 1.5    | ↑ 5.6   |

the scene, demonstrating the effectiveness of the proposed reconstruction-based spatial tokens.

We further evaluate our model on out-of-distribution benchmarks, as reported in Table 2. Specifically, we compare the SenseNova-SI-1.3 checkpoint with our model after Stages 1 & 2. We exclude the Stage 3 model from this comparison as the reward functions are designed for VSI-Bench and are therefore not appropriate for evaluating generalization to these benchmarks. On BLINK, a comprehensive

<sup>1</sup>SenseNova team reports 67.8 overall, without reporting task results. We re-evaluated the checkpoint, noting that minor discrepancies may occur due to different dependencies and GPU hardware.

benchmark for visual perception and spatial reasoning, our model achieves a 0.5 percentage-point improvement, while exhibiting a modest 1.5-point decrease on CV-Bench. It is worth noting that our model is trained to process 32 input images and perform reasoning over six selected frames, whereas BLINK and CV-Bench typically contain substantially fewer images. To accommodate examples with fewer than six frames, we use a balanced frame allocation strategy where available frames are repeated as necessary to preserve the expected input structure.

We also evaluate our model on NExT-QA, a general video question-answering benchmark, and observe a notable 5.6 percentage-point improvement. This result suggests that the enhanced spatial representations and reasoning capabilities may also benefit more general video understanding.

## Qualitative Results

To show the visual verifiability of our method, Fig. 4 presents several examples of reconstructed 3D scenes and predicted 3D object bounding boxes. Each modality is decoded from a dedicated subset of the VLM’s latent states using lightweight projection, cross-attention and MLP layers. Unlike specialized 3D reconstruction or object-prediction methods, which typically rely on large models trained on extensive datasets, our decoders are trained with comparatively limited supervision. We therefore do not expect perfectly accurate reconstructions. Nevertheless, these visualizations confirm that the

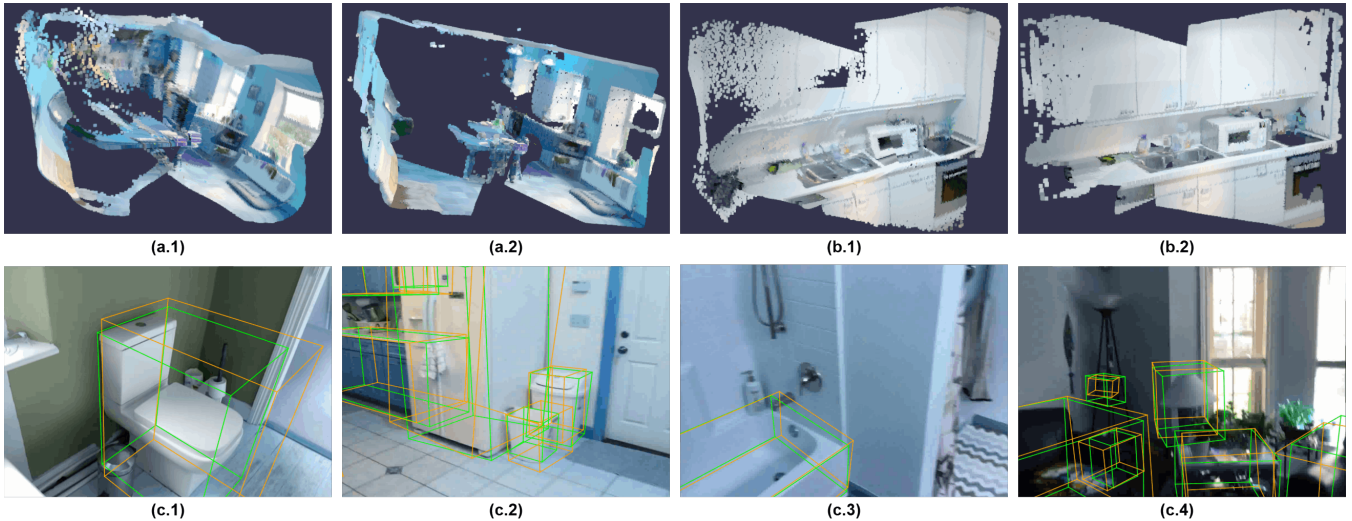


Figure 4: Qualitative comparison of generated 3D scenes (Top) and predicted 3D object bounding boxes (Bottom). (a.1) and (b.1) showcase the decoded 3D token predictions, while (a.2) and (b.2) depict the corresponding expert predictions from VGGT-Omega. In the bottom images, (c.1-4), the orange lines show the prediction and the green lines show the ground truth.

Table 3: Ablation study on data scaling. Additional training on VICA-322K alone degrades performance, whereas Space Tokens improves VSI-Bench average score after SFT and RL fine-tuning.

| Method                                       | Data Samples (Million) | VSI-Bench Avg. |
|----------------------------------------------|------------------------|----------------|
| Qwen3-VL-8B                                  | —                      | 59.8           |
| Qwen3-VL-8B + SenseNova-SI                   | 14M                    | 67.6           |
| Qwen3-VL-8B + SenseNova-SI + VICA-322K       | 14.3M                  | 67.1           |
| Qwen3-VL-8B + SenseNova-SI + stage 1 & 2     | 14.6M                  | 68.6           |
| Qwen3-VL-8B + SenseNova-SI + stage 1 & 2 & 3 | 14.6M                  | <b>68.9</b>    |

learned latent representations capture meaningful geometric information, while our quantitative results demonstrate their effectiveness in improving VQA reasoning.

## Ablations

**Data Scaling Ablations** To isolate the contribution of the proposed tokens, we additionally fine-tune SenseNova-SI-1.3 on the same VICA-322K dataset without introducing our Space Tokens training pipeline. The results are given in Table 3; this baseline achieves a lower overall average score on VSI-Bench compared to the pre-trained SenseNova-SI-1.3 baseline, indicating that the model may be overfitting to the VICA-322K dataset, as this data is already present in SenseNova-SI. However, after training on the same dataset with Space Tokens, the results surpass the SenseNova-SI-1.3 checkpoint. This validates that the observed improvements in the other experiments arise from the proposed spatial tokens rather than additional data scaling.

**Spatial Task Ablations** We evaluate the contribution of each spatial modality by training all variants on the full VICA-322K dataset in Stage 1, followed by 40K samples in Stage 2 to reduce computational cost. As shown in Table 4, removing either modality decreases performance, while combining scene-level 3D reconstruction and object-level 3D

Table 4: Ablation study on the impact of specific spatial token task representations on the VSI-Bench full split.

| Box Tokens | 3D Tokens | VSI-Bench Score |
|------------|-----------|-----------------|
| ✓          | ×         | 67.8            |
| ×          | ✓         | 67.4            |
| ✓          | ✓         | <b>68.4</b>     |

bounding-box supervision achieves the best result on VSI-Bench. Because Stage 1 focuses only on learning the spatial tokens, the model’s general reasoning ability temporarily degrades, and the limited Stage 2 training does not fully recover the original SenseNova-SI-1.3 performance. Nevertheless, the results clearly show that both spatial modalities contribute positively to the model’s reasoning ability.

## Conclusion

We introduce Space Tokens, a lightweight framework that integrates scene-level geometry and object-centric spatial information into VLMs through continuous latent tokens. Our method enables the model to learn rich spatial priors without architectural modifications. Unlike approaches that only evaluate spatial understanding through downstream performance, Space Tokens produces verifiable representations that encode meaningful geometric information and support reasoning. Experiments on VSI-Bench and OOD benchmarks demonstrate consistent gains over pretrained baselines, including state-of-the-art performance on geometry-intensive tasks such as object- and room-size estimation. Ablations further show that these improvements arise from the proposed spatial representations rather than additional data scaling. More broadly, Space Tokens provides a general interface for incorporating interpretable continuous modalities into chain-of-thought reasoning beyond 3D geometry.

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## A. Methodological Details

### A.1. Detailed 3D Reconstruction Formulation

To compute the point cloud reconstruction loss,  $\mathcal{L}_{pt}$ , we first convert both the predicted depth map,  $\hat{D}_i$ , and the teacher depth map,  $D_i$ , into point maps. A point map,  $P$ , is obtained by back-projecting each depth pixel into the camera coordinate frame using the corresponding camera intrinsics. For a pixel at image coordinates  $(u, v)$  with depth value  $z = D_i^{(u,v)}$ , the corresponding 3D point is given by

$$\mathbf{p}_i^{(u,v)} = \begin{bmatrix} \frac{(u-c_x)z}{f_x} & \frac{(v-c_y)z}{f_y} & z \end{bmatrix}^T, \quad (\text{S1})$$

where  $(c_x, c_y)$  denotes the camera principal point and  $(f_x, f_y)$  are the focal lengths, all of which are provided by the camera geometry  $\mathbf{g}_i$ . Applying this transformation to every pixel produces the point map used to compute  $\mathcal{L}_{pt}$ .

In practice, computing teacher depth predictions for all 32 input images simultaneously is memory intensive during Stage 1 training. Instead, we randomly sample a smaller contiguous window of frames from each sequence and generate teacher depth maps only for this subset. The corresponding subset of 3D tokens is then decoded to produce the predicted depth maps,  $\hat{D}_i$ , ensuring that the reconstruction losses are evaluated only on the sampled frames. During Stage 2, we follow the same procedure, except that the sampled frames correspond to the sparse set of images selected for spatial reasoning rather than a randomly chosen dense temporal window.

All loss terms in  $\mathcal{L}_{3D}$  are assigned equal weight. We found that setting every balancing coefficient  $\lambda$  to 1 provided stable optimization while avoiding the need for additional hyperparameter tuning, and therefore use this configuration for all experiments.

### A.2. Detailed Object 3D Bounding Box Generation

Given the LLM hidden states associated with the bounding-box tokens, denoted by  $H_v$ , we project them into a (d)-dimensional embedding space and normalize the resulting features along the feature dimension:

$$\bar{e}_v = \frac{H_v W_{\text{proj}}^\top}{\|H_v W_{\text{proj}}^\top\|_2} \in \mathbb{R}^{T \times d} \quad (\text{S2})$$

We subsequently apply cross-attention between a set of learnable object queries,  $Q$ , and the normalized token representations:

$$a_v = \text{MHA}_{\text{cross-attn}}(Q, \bar{e}_v, \bar{e}_v), \quad a_v \in \mathbb{R}^{T \times d} \quad (\text{S3})$$

The  $Q$  is identical for every view and is a fixed bank of learned slot queries, each notionally responsible for one candidate object. Next, the attended token features are augmented with embeddings of the normalized 2D object bounding boxes:

$$s_v = a_v + \lambda_{\text{bb}} B_v W_{\text{bbox}}^\top, \quad \lambda_{\text{bb}} = 0.2 \quad (\text{S4})$$

Here,  $B_v$  denotes the normalized 2D bounding-box coordinates, and  $W_{\text{bbox}}$  projects them into the same feature space as  $a_v$ . This residual conditioning provides explicit object-localization cues for 3D bounding-box prediction. To further incorporate fine-grained visual information, the visual features  $X_v$  produced by the VLM’s vision backbone are first projected into the decoder embedding space and then normalized:

$$\tilde{X}_v = X_v W_{\text{vis}}^\top \quad (\text{S5})$$

$$\bar{X}_v = \frac{\tilde{X}_v}{\|\tilde{X}_v\|_2} \quad (\text{S6})$$

We use the object-conditioned representations  $s_v$  as queries in a second cross-attention operation over the visual features:

$$g_v = \text{MHA}_{\text{slot-vis-attn}}(s_v, \bar{X}_v, \bar{X}_v) \quad (\text{S7})$$

The resulting visual features  $g_v$  are concatenated with the object-conditioned token features  $s_v$ , followed by layer normalization, linear projection, and a GELU activation:

$$f_v = \text{GELU}(\text{LN}([s_v \| g_v]) W_2^\top) \in \mathbb{R}^{T \times 1024}, \quad (\text{S8})$$

Finally, a linear prediction head maps the fused representations to the 3D bounding-box parameters:

$$z_v = f_v W_{\text{head}}^\top \in \mathbb{R}^{T \times 12} \quad (\text{S9})$$

We represent each predicted 3D bounding box as  $\mathbf{p}_{vi} \in \mathbb{R}^{12}$ , containing its camera-space center, dimensions, and 6D rotation representation. For view  $v$ , the  $S$  prediction slots are matched to  $K_v$  ground-truth boxes using Hungarian assignment:

$$\pi_v^* = \arg \min_{\pi \in \mathfrak{S}_S} \sum_{i=1}^S C_{vi, \pi(i)}, \quad (\text{S10})$$

where

$$C_{vij} = \begin{cases} \frac{1}{12} \|\mathbf{p}_{vi} - \mathbf{g}_{vj}\|_1, & j \leq K_v, \\ \frac{1}{12} \|\mathbf{p}_{vi}\|_1, & j > K_v. \end{cases} \quad (\text{S11})$$

Columns  $j > K_v$  correspond to background targets. After matching, we use the Smooth  $\ell_1$  penalty

$$\rho_\beta(r) = \begin{cases} \frac{r^2}{2\beta}, & |r| < \beta, \\ |r| - \frac{\beta}{2}, & |r| \geq \beta, \end{cases} \quad \beta = 0.1. \quad (\text{S12})$$

The loss for prediction slot  $i$  in view  $v$  is

$$\ell_{vi} = w_{vi} \frac{1}{12} \sum_{d=1}^{12} \rho_{0.1}(p_{vid} - q_{vid}), \quad (\text{S13})$$

where the target and weight are defined as

$$(\mathbf{q}_{vi}, w_{vi}) = \begin{cases} (\mathbf{g}_{v, \pi_v^*(i)}, 1.0), & \pi_v^*(i) \leq K_v, \\ (\mathbf{0}, 0.35), & \pi_v^*(i) > K_v. \end{cases} \quad (\text{S14})$$



Table S1: Space Tokens Stage 1 &amp; 2 Hyperparameters

| Hyperparameter              | Value |
|-----------------------------|-------|
| LoRA rank                   | 16    |
| LoRA alpha                  | 32    |
| LoRA dropout                | 0.01  |
| Learning rate               | 5e-5  |
| Adam $\beta_1$              | 0.9   |
| Adam $\beta_2$              | 0.999 |
| Warmup percentage           | 0.05  |
| Weight decay                | 0.1   |
| Batch size                  | 1     |
| Gradient accumulation steps | 4     |
| Gradient clipping           | 1.0   |

The final 3D bounding-box loss is summed over prediction slots and averaged over annotated views:

$$\mathcal{L}_{3D-BB} = \frac{\sum_v s_v \sum_{i=1}^S \ell_{vi}}{\max\left(1, \sum_v s_v\right)}, \quad (\text{S15})$$

where  $s_v \in \{0, 1\}$  indicates whether view  $v$  has 3D bounding-box supervision.

### A.3. GRPO Reward Design

In stage 3, we design the reward to target the VSI-Bench metric. For multiple choice questions, we extract the ground truth answer and perform fuzzy matching against the predicted choice, if a match occurs then a reward of 1 is assigned. For numeric answer questions we employ the mean relative accuracy (Yang et al. 2025) as a reward. Specifically, for a set of confidence intervals  $\mathcal{C} = \{0.5, 0.55, \dots, 0.95\}$  the reward for prediction  $\hat{y}$  and ground truth answer  $y$  is given as:

$$R_{MRA} = \frac{1}{10} \sum_{\theta \in \mathcal{C}} \mathbb{1} \left( \frac{|\hat{y} - y|}{y} < 1 - \theta \right) \quad (\text{S16})$$

## B. Implementation Details

### B.1. Model Training

For stage 1 & 2 of Space Tokens, we fine-tune the base VLM using low-rank adaption. During fine-tuning, the vision tower and llm backbone are frozen, and only the LoRA parameters are updated. We use the hyperparameter configuration listed in Tab. S1 for these stages. For the GRPO stage, we merge the stage 2 LoRA weights into the VLM and instantiate new LoRA layers. We use the hyperparameter configuration in Tab. S2 for the GRPO stage. We use a random seed of 42

### B.2. Prompting Structure

Different training stages require distinct prompt structures, as each stage emphasizes a different learning objective. As illustrated in Fig. 3, the complete set of spatial tasks is specified in the prompt at every stage. In Stage 1, the prompt contains the

Table S2: Space Tokens Stage 3 Hyperparameters

| Hyperparameter              | Value |
|-----------------------------|-------|
| LoRA rank                   | 16    |
| LoRA alpha                  | 32    |
| LoRA dropout                | 0.01  |
| Learning rate               | 1e-6  |
| Adam $\beta_1$              | 0.9   |
| Adam $\beta_2$              | 0.999 |
| Warmup percentage           | 0.1   |
| Weight decay                | 0.1   |
| Batch size                  | 1     |
| Group size                  | 8     |
| Temperature                 | 1.0   |
| KL $\beta$                  | 0.04  |
| Gradient accumulation steps | 4     |
| Gradient clipping           | 1.0   |

visual inputs together with an instruction to estimate the enabled spatial tasks. Structured task-token blocks are inserted into the assistant context, and the model is optimized exclusively using the auxiliary spatial-alignment losses, without language-model supervision.

In Stages 2 and 3, we introduce a structured, per-view chain-of-thought enclosed within `<think>...</think>` tags, which associates each selected image with its corresponding spatial-task tokens. By default, this reasoning sequence is provided through teacher forcing, and the model is trained to predict only the final answer. Alternatively, the reasoning trace can be included in the prediction target, allowing the model to generate it autoregressively. However, we find that explicitly generating the reasoning increases inference time without providing a meaningful performance improvement over teacher-forced reasoning.

The reasoning structure in Stages 2 and 3 uses only a subset of the input frames rather than all 32 frames. To identify the most informative views, we first trained a model to generate the reasoning trace and used CLIP similarities between the textual prompt and the 32 input frames to rank the frames by relevance. We additionally observed that, when answering questions from our primary benchmark, VSI-Bench, the trained model relied more frequently on certain frames than on others. Based on this observation, we selected the six highest-ranked frames (1, 10, 12, 15, 21, and 28) and used only these views for spatial reasoning during Stages 2 and 3. As illustrated in Fig. S1, the reasoning section of the prompt begins with the statement *The most informative frames are image 1, image 10, ..., followed by reasoning over the corresponding 3D scene representations and 3D object bounding boxes.*

We further analyze the effect of the number of reasoning frames using the model obtained after Stages 1 and 2. As shown in Tab. S3, reasoning over all 32 frames achieves performance comparable to the original SenseNova-SI-1.3 checkpoint, which does not perform explicit spatial reasoning and therefore corresponds to zero reasoning frames. In contrast, using six selected frames yields the best performance

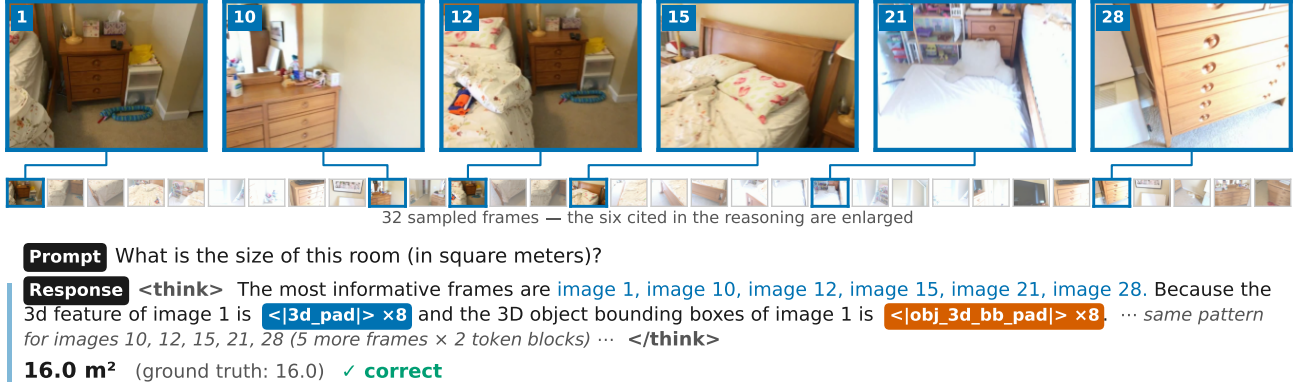


Figure S1: Detailed prompt structure for stage 2 and 3 training. In this prompt style, we perform the reasoning based on six pre-selected frames and generate the answer.

Table S3: Comparison between number of frames for reasoning in the prompt

| Number of Frames for Reasoning | VSI-Bench Avg. |
|--------------------------------|----------------|
| 0 (Sensenova1.3)               | 67.6           |
| 32                             | 67.6           |
| 4                              | 68.4           |
| 6                              | 68.6           |

among the evaluated configurations. These results suggest that reasoning over a compact set of informative views is more effective than processing the entire frame sequence indiscriminately. Many frames may depict uninformative or question-irrelevant regions, such as ceilings, the ground, or unrelated parts of the scene, thereby introducing noise into the reasoning process. Consequently, selecting a subset of relevant frames provides a more focused and effective basis for spatial reasoning, consistent with both intuition and our empirical findings.

## C. Additional Ablations

### C.1. Token Replacement Test

To evaluate the utility of newly introduced tokens, (Zhang, Bigverdi, and Krishna 2026) proposed the Token Replacement Test (TRT), in which the input embeddings corresponding to the proposed tokens are perturbed before inference. The perturbation is done by replacing the token embedding with either a zero vector, a random vector, or a random vector sampled from the same distribution as the latent representation. If downstream performance deteriorates under this perturbation, the test suggests that the model relies on those token embeddings when producing its predictions.

While TRT is a useful diagnostic for methods trained on single-image inputs, such as CoVT (Qin et al. 2025), its interpretation becomes less clear in multi-image settings. In single-image models, each learned token has a fixed semantic

Table S4: Token replacement test results on VSI-Bench. Perturbing spatial token input embeddings negligibly impacts spatial understanding.

| Mode                | VSI-Bench Avg. |
|---------------------|----------------|
| zero                | 68.82          |
| random              | 68.55          |
| random (same dist.) | 68.69          |

role and its input embedding can directly indicate where the model should extract information. In contrast, multi-image models must additionally identify which image each token corresponds to. This association cannot be inferred from a constant input embedding alone, since the same learned token is reused across every image. Instead, image identity is specified through positional information elsewhere in the prompt.

For Space Tokens, this positional information is explicitly encoded in the reasoning template. For example, the phrase “Because the 3D feature of image 5 is <|3d\_pad|> ×8” informs the model that the subsequent spatial tokens correspond to the fifth input image. Consequently, the input embedding primarily identifies the token type, while the prompt structure determines which image-specific latent representation should be retrieved.

This distinction suggests that TRT primarily measures the model’s sensitivity to perturbations of the *input embedding*, rather than the usefulness of the latent representations computed from those tokens. Consistent with this interpretation, Mull-Tokens (Ray et al. 2026), which is also trained on multi-frame visual inputs, reports that TRT produces only negligible changes in performance on BLINK (Fu et al. 2024). We observe the same behavior for our Space Tokens SenseNova-SI-1.3 Stage 3 model when evaluated on VSI-Bench, as shown in Tab. S4. The minimal performance degradation therefore should not be interpreted as evidence that the spatial token representations are unused; rather, it in-

Table S5: Attention masking results on VSI Bench. Removing attention from spatial tokens greatly impacts spatial understanding.

| Mode                  | Avg. | Numerical Questions |            |           |           | Multiple-Choice Questions |           |            |             |
|-----------------------|------|---------------------|------------|-----------|-----------|---------------------------|-----------|------------|-------------|
|                       |      | Obj. Cnt.           | Abs. Dist. | Obj. Size | Room Size | Rel. Dist.                | Rel. Dir. | Route Plan | Appr. Order |
| No masking            | 63.4 | 72.3                | 55.8       | 79.0      | 63.2      | 62.0                      | 71.9      | 30.9       | 72.2        |
| Spatial token masking | 55.3 | 60.4                | 46.3       | 63.7      | 58.2      | 50.8                      | 60.1      | 32.0       | 70.6        |

icates that perturbing the input embeddings is an insufficient probe for evaluating latent representations in multi-image architectures.

To directly measure whether the information encoded in the spatial token representations contributes to model predictions, we instead propose a causal attention masking experiment.

## C.2. Token Attention Masking

Rather than perturbing the token embeddings before they are processed by the transformer, we instead intervene directly on the resulting latent representations. Specifically, we construct a causal attention mask that prevents the VLM from attending to the spatial tokens during answer generation while leaving all other computations unchanged.

The results of attention masking on the Space Tokens SenseNova-SI-1.3 Stage 3 model are reported in Tab. S5. Removing attention to the spatial tokens produces a substantial decrease in VSI-Bench performance, which is especially notable in the object size metric, which observes a 15.3 point drop. This demonstrates that the information encoded in these latent representations is causally required for accurate spatial reasoning. Unlike TRT, which probes sensitivity to the initial input embeddings, this experiment directly evaluates whether the learned representations themselves are utilized by the VLM.

Since Flash Attention 2 does not support arbitrary attention masks, these experiments are performed using eager attention. It should be noted that eager attention is slightly less efficient and produces a small reduction in absolute performance compared with Flash Attention 2, as our model was trained with Flash Attention 2. However, this effect is common to both the masked and unmasked evaluations and therefore does not affect the conclusions of the study.